# Networking concepts and security

**Prof .Vaishalee Joishar**
CSE-Cyber Security
PIET,Parul University

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# NETWORK PENETRATION TESTING

Network penetration testing, often called "**pentesting**" is a method of evaluating the security of a computer network by safely attempting to exploit vulnerabilities. It involves simulating an attack on a network to identify potential weaknesses that malicious hackers could exploit.Pentesters use various tools and techniques to assess the network's security posture, attempting to gain unauthorized access to systems, applications, or data.

# STEPS FOR NETWORK PENTESTING

- Planning and Preparation
- Reconnaissance
- Vulnerability Analysis
- Exploitation
- Post-Exploitation
- Documentation and Reporting
- Follow-Up

# Planning and Preparation

❖**Define Objectives:** Determine the scope and goals of the penetration test. Identify what systems, applications, or assets will be tested.

❖**Gather Information:** Collect details about the network architecture, IP ranges, systems, applications, and potential entry points.

❖**Get Permissions:** Ensure you have explicit permission from the network owner to perform the test to avoid legal issues.5

# Reconnaissance

❖**Passive Reconnaissance:** Gather publicly available information about the target network using non-intrusive methods such as searching online, social media, DNS lookups, etc.

Tools : WHOIS, Netcraft, Shodan, Wapplyzer,NSLOOKUP

❖**Active Reconnaissance:** Use tools to actively scan the network, identifying live hosts, open ports, services, and operating systems.

Tools : Nmap, Nikto, Nessus, OpenVAS, AngryIPScanner

# Tools for Reconnaisance:WHOIS

❖**ICANN,** the International Corporation for Assigned Names and Numbers, regulates the WHOIS database.

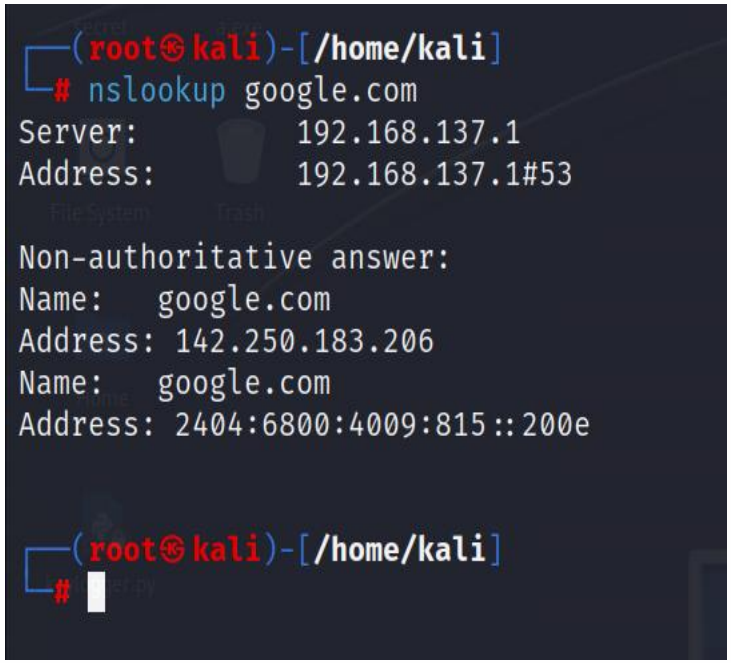
❖**WHOIS** is a public database that houses the information collected when someone registers a domain name or updates their DNS Settings.

# WHOIS

Free Whois Lookup - Whois IP S
https://www.whois.com/wh
Getting Started
New Tab
Learning on Simplilearn
Other Bookmarks
Whois
Identity for everyone
Whois Domain Lookup
Whois search for Domain and IP
google.com
SEARCH
Example: qq.com, google.co.in, bbc.co.uk, ebay.ca

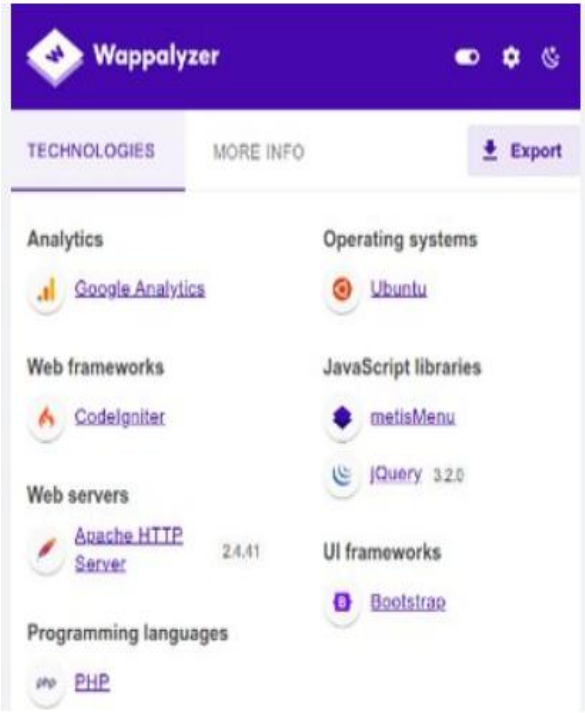
# Passive Reconnaissance : NSLOOKUP

nslookup is a network administration command-line tool for querying the Domain Name System to obtain the mapping between domain name and IP address, or other DNS records



# Passive Reconnaissance : WAPPLYZER

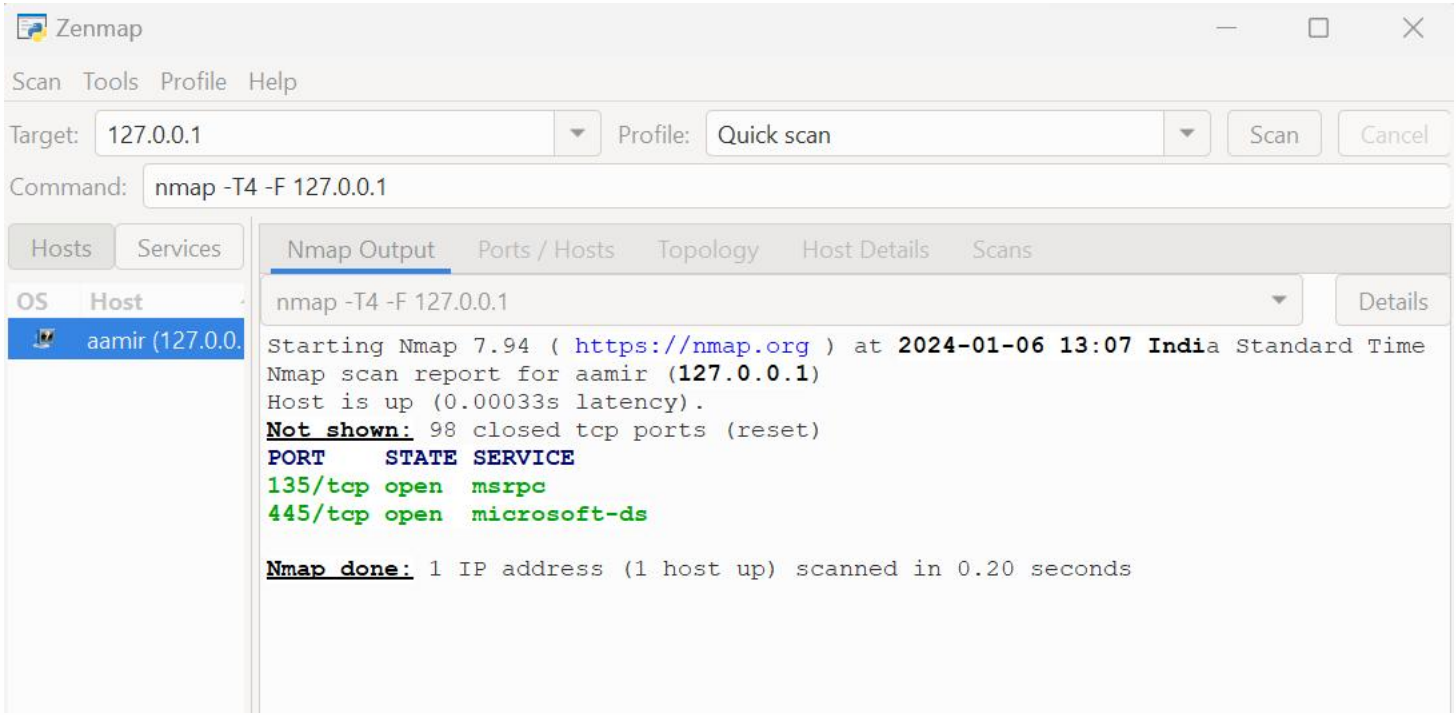
Wappalyzer is more than a CMS detector or framework detector: it **uncovers more than a thousand technologies in dozens of categories such as programming languages, analytics, marketing tools, payment processors, CRM, CDN and others**.



# Host Discovery: Nmap/Zenmap

Nmap (Network Mapper) is a versatile and powerful open-source tool used for network discovery, security scanning, and auditing. When used for network reconnaissance, Nmap provides a wealth of information about the target network about open ports, Service Version Detection, OS Fingerprinting, Network Topology etc.

# Nmap/Zenmap

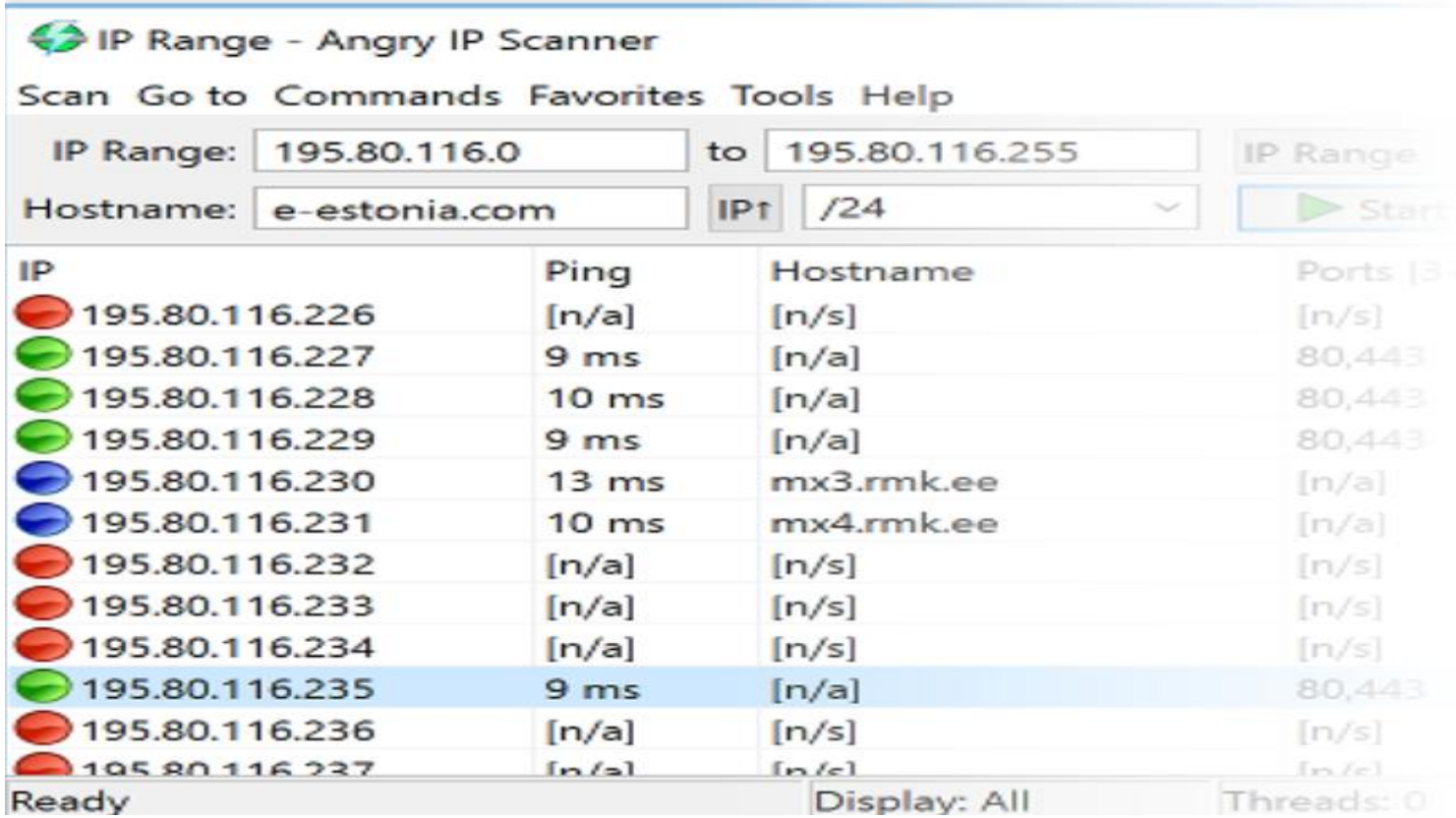

Zenmap
Scan Tools Profile Help
Target: 127.0.0.1
Profile: Quick scan
Scan
Cancel
Command: nmap -T4 -F 127.0.0.1
Hosts
Services
OS Host
aamir (127.0.0.
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -T4 -F 127.0.0.1
Details
Starting Nmap 7.94 ( https://nmap.org ) at 2024-01-06 13:07 India Standard Time
Nmap scan report for aamir (127.0.0.1)
Host is up (0.00033s latency).
Not shown: 98 closed tcp ports (reset)
PORT STATE SERVICE
135/tcp open msrpc
445/tcp open microsoft-ds
Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds

# Host Discovery: Angry IPScanner

Angry IP Scanner (or simply ipscan) is an open-source and cross-platform network scanner designed to be fast and simple to use. It scans IP addresses and ports as well as has many other features.

It is widely used by network administrators and just curious users around the world, including large and small enterprises, banks, and government agencies.

# Host Discovery : Angry IPScanner



# Open Port/Service Identification

- Attackers can use information about open ports to target specific services or applications that may have known vulnerabilities. Open ports may serve as entry points for exploitation.
- Knowing the versions of services running on open ports helps attackers identify known vulnerabilities associated with specific software versions. They can then launch targeted attacks against these vulnerabilities.
- Detecting the underlying operating system of a target system can help attackers tailor their exploits to vulnerabilities commonly found in that OS. It aids in crafting more effective and targeted attacks.

# Banner Grabbing/OS Version Check

**Banner Grabbing:**

- **Banner grabbing is a technique used to gain information about a computer system on a network and the services running on its open ports.**
- **Administrators can use this to take inventory of the systems and services on their network.**
- **Tools commonly used to perform banner grabbing are Telnet, nmap, zmap and Netcat.**

# Vulnerability Analysis

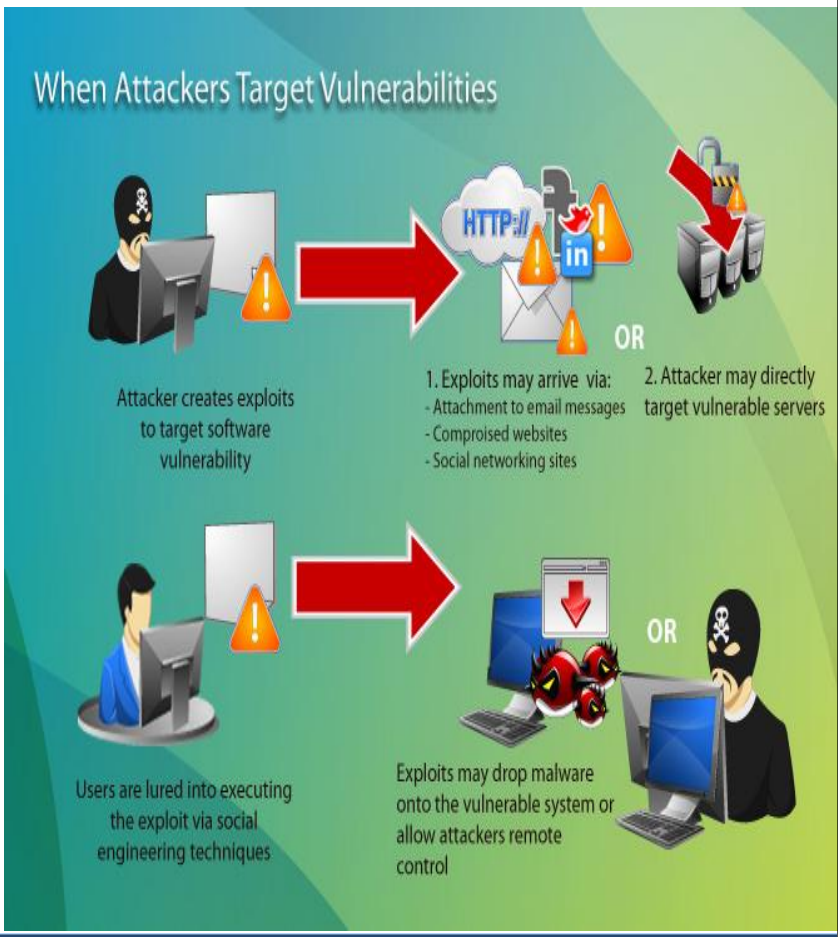
❖**Vulnerability Scanning:** Use automated tools to scan the network for known vulnerabilities in systems, services, and applications.

❖**Manual Analysis:** Manually inspect systems and services for potential weaknesses that automated tools might miss, such as misconfigurations or logic flaws.

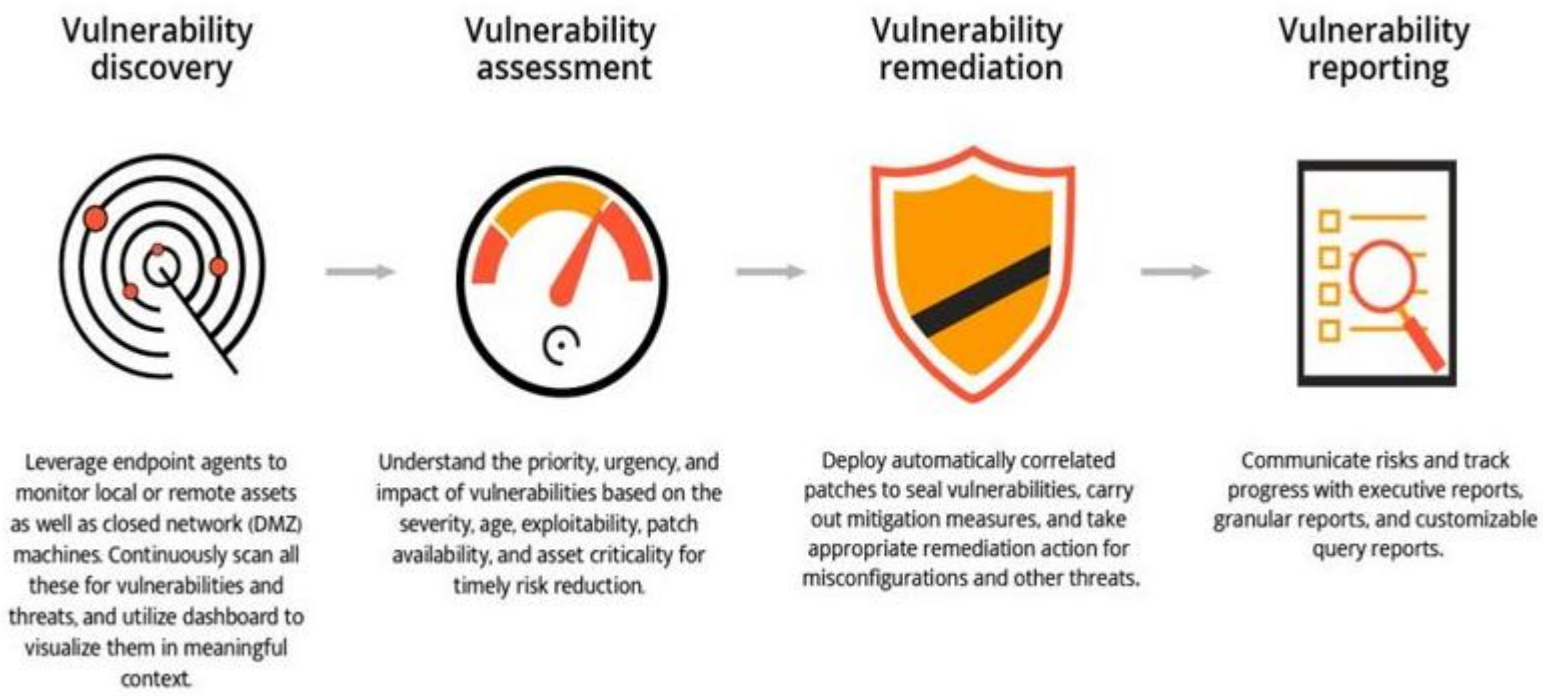
# Overview of Vulnerability Scanning

**Vulnerability**

- **Vulnerability is a weakness which allows an attacker to reduce a system's security.**
- **Vulnerability scanning usually refers to the scanning of systems that are connected to the Internet.**
- **It can also refer to system scanning or audits on internal networks that are not connected to the Internet in order to assess the threat of malicious software.**
- **It is possible to know the basic security measures when installing and managing network and websites. but it is not possible to catch all the vulnerabilities reside in the network and websites**



# Contd.....


Vulnerability discovery
Vulnerability assessment
Vulnerability remediation
Vulnerability reporting
Leverage endpoint agents to monitor local or remote assets as well as closed network (DMZ) machines. Continuously scan all these for vulnerabilities and threats, and utilize dashboard to visualize them in meaningful context.
Understand the priority, urgency, and impact of vulnerabilities based on the severity, age, exploitability, patch availability, and asset criticality for timely risk reduction.
Deploy automatically correlated patches to seal vulnerabilities, carry out mitigation measures, and take appropriate remediation action for misconfigurations and other threats.
Communicate risks and track progress with executive reports, granular reports, and customizable query reports.

# Classifications Of Vulnerability

**Vulnerability originates from three sources:**

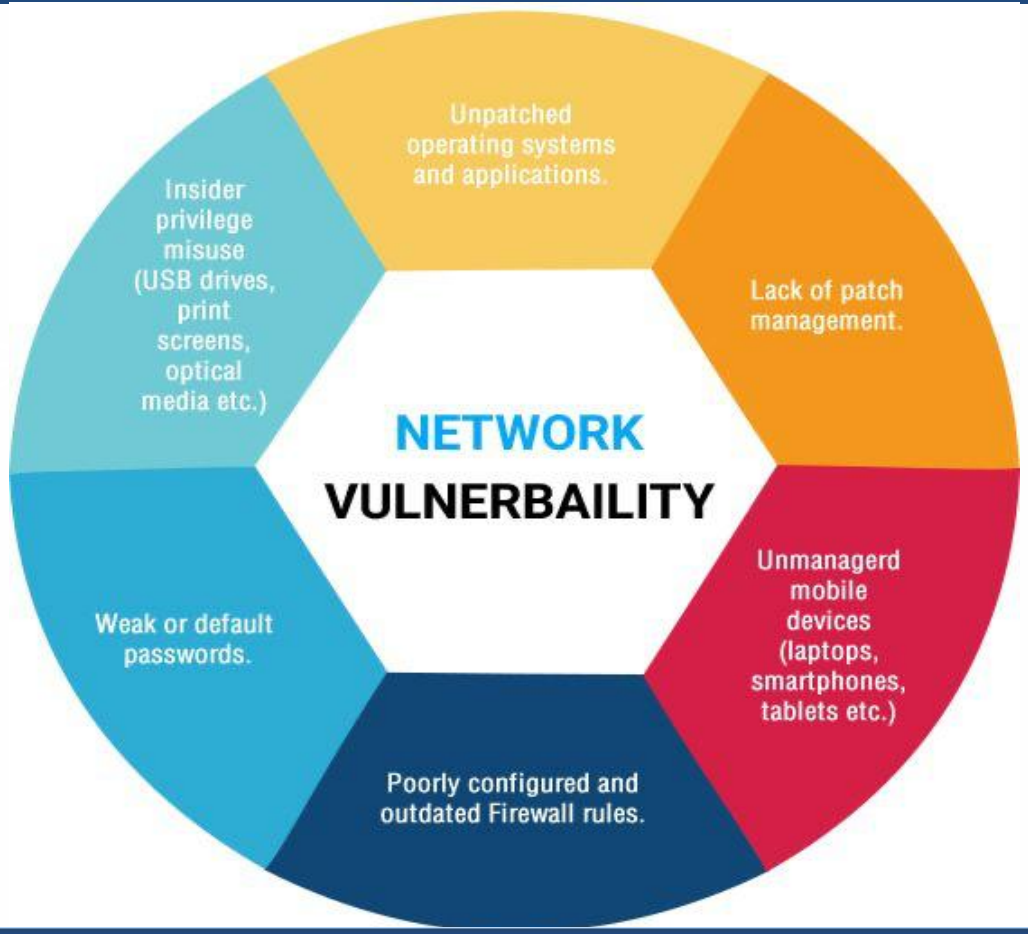
- Vendor-originated: This includes software bugs, missing operating system patches, vulnerable services, insecure default configurations, and web application vulnerabilities.

- System administration-originated: This includes incorrect or unauthorized system configuration changes, lack of password protection policies, and so on.

- User-originated:This includes sharing directories to unauthorized parties, failure to run virus scanning software, and malicious activities, such as deliberately introducing system backdoors.

# Contd.....

- The vulnerability scanners provide you the automate security auditing and play an important role in your IT security.
- The vulnerability scanners can scan your network and websites for up to thousands of different security risks.
- It produces a list of those vulnerabilities, and gives steps on how to overcome or reduce them.



# Types Of Vulnerability Scanners

**Network Vulnerability Scanners:** These tools scan network infrastructure, devices, and systems for weaknesses, misconfigurations, or potential entry points for attackers.
Tools: Nessus, OpenVAS, and Rapid7 Nexpose.

**Web Application Scanners:** Focused on web applications, these scanners detect vulnerabilities like SQL injection, cross-site scripting (XSS), and other web-specific issues.
Tools :Acunetix, OWASP ZAP, and BurpSuite.

# Types Of Vulnerability Scanners

**Host-Based Scanners:** These scanners assess vulnerabilities on individual systems (servers, workstations) by analyzing software configurations, patch levels, and known security issues. Tools like Qualys, Tripwire, and OpenVAS offer host-based scanning capabilities.

**Database Scanners:** Specifically designed to identify weaknesses within databases, these scanners detect misconfigurations, access control issues, and vulnerabilities in database management systems. Examples include AppDetectivePRO and IBM Security Guardium.

# Types Of Vulnerability Scanners

**Mobile Application Scanners:** Geared towards mobile app security, these tools scrutinize mobile applications for vulnerabilities that might compromise user data or expose the app to attacks.

**Tools :MobSF (Mobile Security Framework) and QARK (Quick Android Review Kit)**

**Cloud-Based Vulnerability Scanners:** Tailored for cloud environments, these scanners assess cloud infrastructure for misconfigurations, security gaps, and compliance issues specific to cloud platforms like AWS, Azure, or Google Cloud.

**Tools: AWS Inspector and CloudSploit**

# Types Of Vulnerability Scanners

**Passive Scanners:** Unlike active scanners that actively send traffic to systems to identify vulnerabilities, passive scanners monitor network traffic and system behavior to identify potential weaknesses without interacting directly with the target. Security Information and Event Management (SIEM) tools often include passive scanning capabilities.

# False Negative

A false negative occurs when a test or security system fails to detect something that is actually present or true. In cybersecurity, this means the system wrongly identifies a vulnerability or threat as safe or non-existent. It's a failure to recognize an actual issue, potentially leading to a security breach if the vulnerability isn't addressed. For instance, a vulnerability scanner might miss a known security flaw in a system, falsely indicating that the system is secure.

# False Positive

On the other hand, a false positive happens when a test or security system identifies something as true or present when it's actually not. In cybersecurity, this means the system flags something as a vulnerability or threat when, in fact, it's not a genuine risk. False positives can waste time and resources by prompting unnecessary investigations or actions. For example, a vulnerability scanner might mistake a benign system configuration for a security flaw, triggering an alert.

# Exploitation

❖**Exploit Vulnerabilities:** Attempt to exploit identified vulnerabilities to gain unauthorized access. This might involve using various techniques such as password cracking, SQL injection, buffer overflow attacks, etc.

❖**Privilege Escalation:** Once access is gained, attempt to escalate privileges to gain higher levels of access within the network.

# Common Network Vulnerabilities

**Open Ports and Services:**

**Description:** Unnecessary open ports or services can provide attackers with entry points into a network.

**Prevention:** Regularly scan and close unused ports, disable unnecessary services, and employ firewalls to restrict access.

**Weak Passwords:**

**Description:** Default or easily guessable passwords can lead to unauthorized access.

**Prevention:** Enforce strong password policies, implement multi-factor authentication (MFA), and regularly audit and update credentials.

# Common Network Vulnerabilities

**Misconfigured Firewalls:**

**Description:** Improperly configured firewalls may allow unauthorized access or block essential services.

**Prevention:** Regularly review and update firewall rules, restrict unnecessary traffic, and implement logging for firewall events.

**Outdated Software:**

**Description:** Unpatched systems and software can contain known vulnerabilities that attackers can exploit.

**Prevention:** Implement a robust patch management system, regularly update software, and retire end-of-life applications.

# Common Network Vulnerabilities

**Insecure Wireless Networks:**

**Description:** Weakly secured Wi-Fi networks can be exploited for unauthorized access.

**Prevention:** Use strong encryption (WPA3), change default credentials, and implement network segmentation.

**Network Sniffing:**

**Description:** Unauthorized interception of network traffic can lead to the compromise of sensitive data.

**Prevention:** Encrypt sensitive data in transit, use protocols with built-in security (e.g., HTTPS), and monitor for unusual network activity.

# Common Network Vulnerabilities

**Man-in-the-Middle Attacks:**

**Description:** Attackers intercept and manipulate communication between two parties.

**Prevention:** Use strong encryption (TLS/SSL), employ secure key exchange protocols, and regularly monitor for suspicious activities.

**Denial-of-Service (DoS) Attacks:**

**Description:** Overloading resources to disrupt normal network operation.

**Prevention:** Implement network redundancy, use traffic filtering, and employ intrusion prevention systems (IPS) to mitigate DoS attacks.

# Common Network Vulnerabilities

**Vulnerabilities in Network Protocols:**

**Description:** Exploitable weaknesses in communication protocols.

**Prevention:** Keep protocols updated, disable unnecessary protocols, and use secure alternatives when possible.

**Web Application Vulnerabilities:**

**Description:** Flaws in web applications that can be exploited for unauthorized access or data manipulation.

**Prevention:** Regularly update and patch web applications, conduct security assessments, and implement secure coding practices.

# Common Network Vulnerabilities

**Physical Security Weaknesses:**

**Description:** Lack of physical security measures may lead to unauthorized access to network infrastructure.

**Prevention:** Implement access controls, surveillance systems, and physical access restrictions.

**Social Engineering:**

**Description:** Manipulating individuals to disclose sensitive information or perform actions that compromise security.

**Prevention:** Educate users about social engineering tactics, implement user awareness training, and establish a reporting mechanism for suspicious activities.

# Common Network Vulnerabilities

**Insufficient Logging and Monitoring:**

**Description:** Inadequate monitoring can lead to delayed detection of security incidents.

**Prevention:** Implement comprehensive logging, regularly review logs, and use intrusion detection systems (IDS) for real-time monitoring.

**Default Configurations:**

**Description:** Default configurations may have well-known vulnerabilities that attackers can exploit.

**Prevention:** Change default credentials, configure systems securely, and conduct security assessments on default configurations.

# Common Network Vulnerabilities

**Insecure Network Architecture:**

**Description:** Poorly designed network architecture may expose critical systems to unnecessary risks.

**Prevention:** Implement network segmentation, follow the principle of least privilege, and conduct regular security reviews of network architecture.

**VPN Vulnerabilities:**

**Description:** Weaknesses in Virtual Private Network (VPN) configurations may expose communication to interception.

**Prevention:** Use strong encryption algorithms, update VPN software regularly, and monitor VPN connections for anomalies.

# Common Network Vulnerabilities

**IoT Device Vulnerabilities:**

**Description:** Insecure Internet of Things (IoT) devices may introduce vulnerabilities into the network.

**Prevention:** Change default credentials on IoT devices, update firmware regularly, and segment IoT devices from critical network infrastructure.

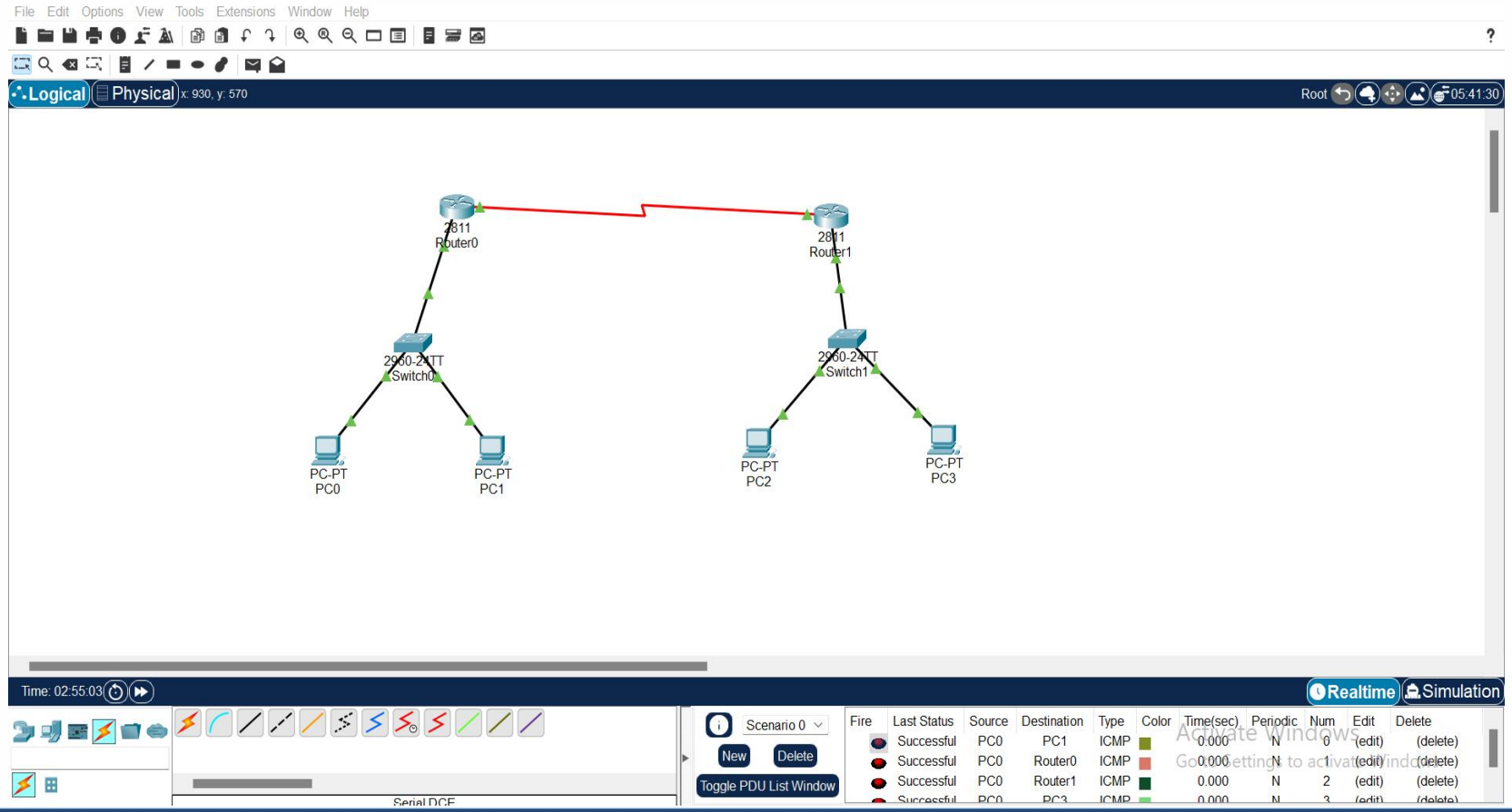
# Zero-day Vulnerability

- Zero-day vulnerability refers to a hole in software that is unknown to the vendor.
- This security hole is then exploited by hackers before the vendor becomes aware and hurries to fix it- this exploit is called a zero day attack.
- Zero-day vulnerabilities are particular dangerous because they represent a gap in knowledge between the attacker and defender.

# Network Diagram

In penetration testing, a network diagram is a visual representation of the target organization's network infrastructure. It illustrates the relationships and connections between various network components, such as servers, routers, switches, firewalls, workstations, and other devices. Network diagrams are valuable tools for penetration testers as they provide a clear overview of the target environment, helping testers understand the network's architecture and identify potential security vulnerabilities.

# Network Diagram



# Prepare Proxy: BurpSuite Proxy

Burp Proxy is a popular web application security testing tool developed by PortSwigger. It is a part of the Burp Suite, which is an integrated platform used for web application security testing. Burp Proxy is specifically designed for intercepting and modifying HTTP and HTTPS traffic between your browser and the target web application. It's commonly employed by security professionals, penetration testers, and developers to identify and fix vulnerabilities in web applications.

# Features of Burp Proxy

**Interception and Modification**: Burp Proxy acts as a middleman between the user's browser and the target web application. It intercepts and allows you to modify the requests and responses flowing between them. This capability is crucial for identifying and manipulating web application vulnerabilities

# Features of Burp Proxy Contd....

Burp Project Intruder Repeater View Help

Dashboard Target Proxy Intruder Repeater Collaborator Sequencer Decoder Comparer Logger Organizer Extensions Learn Settings

**Tasks** New scan New live task

Filter Running Paused Finished Live task Scan Intruder attack

1. Live passive crawl from Proxy (all traffic)

Add links. Add item itself, same domain and URLs in suite scope.

Capturing:

0 items added to site map

0 responses processed

0 responses queued

Time to level up? Catch more bugs with Burp Suite Pro Find out more

**Issue activity [Pro version only]**

Filter Search

| Source | Issue type | Host | Pa |
|---|---|---|---|
| Task 0 | Suspicious input transformation (reflected) | http://insecure-bank.com | /url-shorten |
| Task 0 | SMTP header injection | http://insecure-website.c... | /contact-us |
| Task 0 | Serialized object in HTTP message | http://insecure-bank.com | /blog |
| Task 0 | Cross-site scripting (DOM-based) | https://insecure-bank.com | / |
| Task 0 | XML external entity injection | https://vulnerable-websit... | /product/stock |
| Task 0 | External service interaction (HTTP) | https://insecure-website.... | /product |
| Task 0 | Web cache poisoning | http://insecure-bank.com | /contact-us |
| Task 0 | Server-side template injection | http://insecure-bank.com | /user-homepage |
| Task 0 | SQL injection | https://vulnerable-websit... | / |
| Task 0 | OS command injection | https://insecure-website.... | /feedback/submit |

**Event log**

Filter Critical Error Info Debug Search

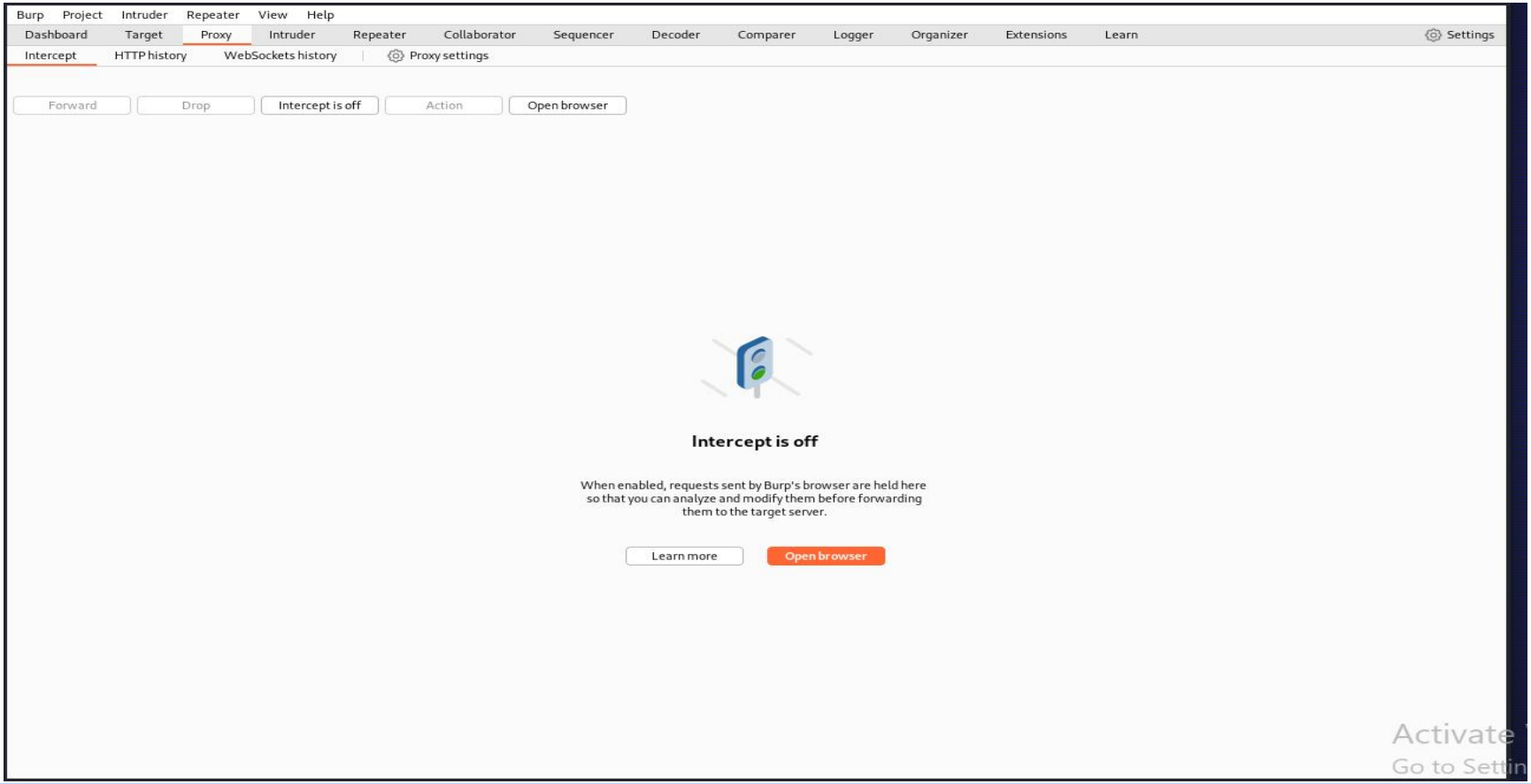
| Time | Type | Source | Message |
|---|---|---|---|
| 15:27:32 19 Jan 2024 | Info | Proxy | Proxy service started on 127.0.0.1:8080 |

Advisory

Memory: 68.6MB Disk: 32KB

Activate
Go to Setti

# Features of Burp Proxy Contd....



# Features of Burp Proxy Contd....

**Request and Response Tampering**: Burp Proxy enables users to modify requests before they reach the server and responses before they reach the browser. This is useful for testing how the application behaves when certain parameters are altered.

# Features of Burp Proxy Contd....

**Decoder Tool**: The Decoder tool in Burp Suite is a utility that assists in manipulating and decoding encoded data, especially during web application security testing. It helps security professionals analyze and understand how data is transformed or encoded before being transmitted over the network.

**Extensions Tool:**The Extensions tool in Burp Proxy allows users to enhance and extend the functionality of Burp Suite by adding custom features, automating tasks, or integrating with external tools. Extensions can be written in Java and can be loaded into Burp Suite to customize and extend its capabilities.

# Features of Burp Proxy Contd....

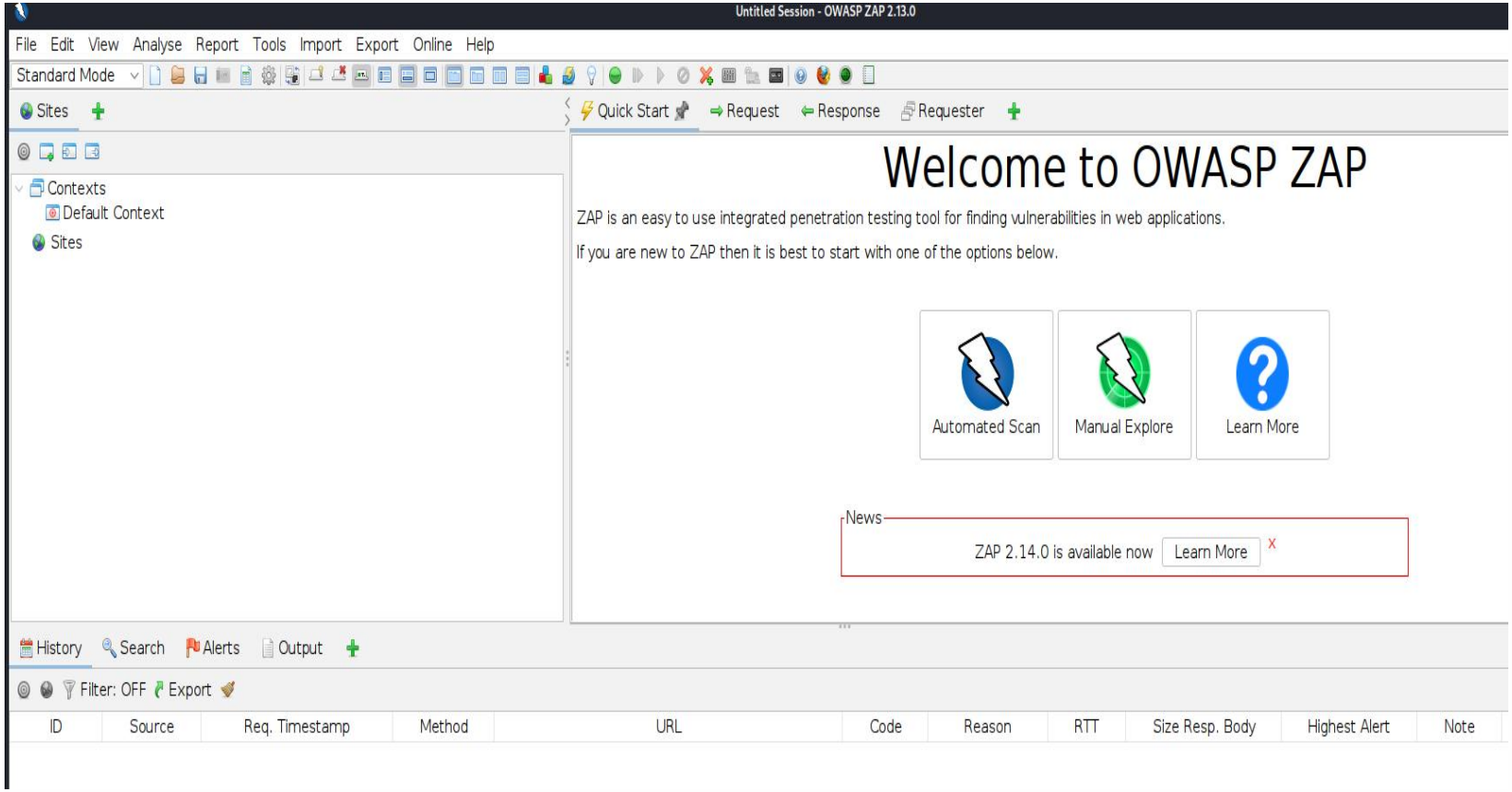
**Repeater Tool**: The Repeater tool in Burp allows users to resend HTTP requests to the server and observe the responses. This is beneficial for testing and modifying specific requests iteratively to analyze application behavior.

**Intruder Tool:**The Intruder tool in Burp Suite is a powerful feature designed for automating customized attacks against web applications. It allows security professionals and penetration testers to perform various types of automated attacks, such as brute force attacks, fuzzing, and payload-based testing, to identify vulnerabilities in a web application.

# OWASP ZAP Proxy

OWASP ZAP (Zed Attack Proxy) is a widely-used open-source security testing tool designed to help find vulnerabilities in web applications during the development and testing phases. ZAP is maintained by the Open Web Application Security Project (OWASP), which is a nonprofit organization focused on improving the security of software.

# OWASP ZAP Proxy



# OWASP ZAP Proxy Features

**Automated Scanning**: ZAP provides automated scanning capabilities to identify common web application vulnerabilities, including but not limited to cross-site scripting (XSS), SQL injection, and security misconfigurations. Users can initiate various types of scans to assess the security posture of their web applications.

# OWASP ZAP Proxy Features

**Manual Testing and Exploration**: ZAP allows security professionals and developers to manually explore and test web applications. The tool includes features for intercepting and modifying HTTP and HTTPS traffic, similar to Burp Proxy. Users can inspect and manipulate requests and responses to identify and understand potential security issues.

# OWASP ZAP Proxy Features

**Active and Passive Scanning**: ZAP offers both active and passive scanning. Active scanning involves actively sending requests to the application to identify vulnerabilities, while passive scanning involves analyzing the traffic and identifying issues without actively sending additional requests.

# OWASP ZAP Proxy Features

**Community-Driven and Open Source**: Being an OWASP project, ZAP is developed and maintained by a community of volunteers. It is open source, making it freely accessible for users to download, use, and contribute to its development.

# Post-Exploitation

- **Maintain Access:** After exploitation, maintain access to systems to assess the potential impact and extent of the compromise.
- **Pivot:** Use the compromised system to gain access to other parts of the network.

# Documentation and Reporting

For each identified vulnerability in a penetration testing report, it's important to provide detailed and comprehensive information to help stakeholders understand the nature of the issue and facilitate effective remediation. Here is a list of key details that should be included for each identified vulnerability:

# Contd…….

**Vulnerability Identifier:**

Assign a unique identifier to each vulnerability for easy reference and cross-referencing within the report.

**Description:**

Provide a clear and concise description of the vulnerability, explaining its nature, how it was discovered, and any potential security implications.

# Contd…….

**Location or Affected Component:**

Specify the location of the vulnerability, including the affected system, network device, or application.

**Risk Level:**

Assign a risk level or severity score using a standardized system (e.g., Common Vulnerability Scoring System – CVSS website: **https://www.first.org/cvss/calculator/3.0)** to indicate the potential impact and exploitability.

# Contd…….

https://www.first.org/cvss/calculator/3.0

Getting Started | New Tab | Learning on Simplilearn | Login | ASPEN | Login - OneLearn Trai... | Download EC-Council ... | OSCP | English Speaking Cour... | Other Bookmarks

FIRST Improving Security Together

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Common Vulnerability Scoring System (CVSS-SIG)

- Calculator
- Specification Document
- User Guide
- Examples
- Frequently Asked Questions
- CVSS v4.0 Documentation & Resources
- CVSS v3.1 Archive
- CVSS v3.0 Archive
  - **CVSS v3.0 Calculator**
  - CVSS v3.0 Specification Document

CVSS

## Common Vulnerability Scoring System Version 3.0 Calculator

Hover over metric group names, metric names and metric values for a summary of the information in the official CVSS v3.0 Specification Document. The Specification is available in the list of links on the left, along with a User Guide providing additional scoring guidance, an Examples document of scored vulnerabilities, and notes on using this calculator (including its design and an XML representation for CVSS v3.0).

# Contd…….

**CVE (Common Vulnerabilities and Exposures) ID:**

If applicable, provide the CVE ID associated with the vulnerability for standardized identification. You can check the CVE-ID on exploit-db website:
**https://www.exploit-db.com/**

**Attack Vector:**

Describe how an attacker could potentially exploit the vulnerability, including details about the attack vector.

# Contd…….

https://www.exploit-db.com

Getting Started | New Tab | Learning on Simplilearn | Login | ASPEN | Login - OneLearn Trai... | Download EC-Council ... | OSCP | English Speaking Cour... | Other Bookmarks

EXPLOIT DATABASE

Verified | Has App | Filters | Reset All

Show 15 | Search: http

| Date | D | A | V | Title | Type | Platform | Author |
|---|---|---|---|---|---|---|---|
| 2023-07-07 | ⬇ | | X | Windows 10 v21H1 - HTTP Protocol Stack Remote Code Execution | Remote | Windows | nu11secur1ty |
| 2023-05-31 | ⬇ | | X | Flexense HTTP Server 10.6.24 - Buffer Overflow (DoS) (Metasploit) | Remote | Multiple | Ege Balci |
| 2023-04-07 | ⬇ | | X | Tenda N300 F3 12.01.01.48 - Malformed HTTP Request Header Processing | Remote | Hardware | @h454nsec |
| 2022-05-11 | ⬇ | | X | Akka HTTP 10.1.14 - Denial of Service | Remote | Multiple | cxosmo |

# Contd…….

**Impact:**

Specify the potential impact of the vulnerability on the confidentiality, integrity, and availability of the affected system or data.

**Evidence and Proof-of-Concept (PoC):**

Include any relevant evidence that supports the existence of the vulnerability, such as screenshots, logs, or data.

If applicable and safe, provide a proof-of-concept to demonstrate the exploitability of the vulnerability.

# Contd…….

**Recommendations:**

Clearly state recommendations for remediation, including step-by-step instructions if possible.

Prioritize recommendations based on risk levels and potential business impact.

**Affected Versions:**

Specify the versions or configurations of software or systems that are vulnerable.

Indicate whether the vulnerability is specific to certain releases.

# Contd…….

**Mitigation Techniques:**

Offer interim mitigation techniques, if available, to reduce the risk while a more permanent fix is being implemented.

**Dependencies:**

Identify any dependencies or prerequisites for remediation, such as addressing other vulnerabilities or implementing specific security controls.

# Contd…….

**Regulatory Compliance Implications:**

If applicable, mention any implications for regulatory compliance and standards that may be affected by the vulnerability.

**Ease of Exploitation:**

Assess and communicate the ease with which an attacker could exploit the vulnerability. This can influence the urgency of remediation.

# Contd…….

**Technical Details:**

Provide technical details about the vulnerability, such as the specific code snippets, configuration settings, or network conditions that contribute to the issue.

**Risk Context:**

Explain the context of the vulnerability within the organization's overall risk landscape. Consider business-critical systems and data.

# Contd…….

**False Positive Confirmation:**

Clearly indicate if the identified vulnerability is later determined to be a false positive, and provide the reasons for the correction.

**Post-Exploitation Details (if applicable):**

Document the steps taken during post-exploitation, including any privilege escalation or lateral movement activities.

# Follow-Up

❖**Discuss Results:** Review the findings with the network owner to ensure a clear understanding of the vulnerabilities and the suggested remediation steps.

❖**Implement Fixes:** Assist in implementing security measures and fixes to address the identified vulnerabilities.